Simulated Teaching Hospital

Compilation in the Style of Photocopied Inpatient Medical Records

Medical record photocopy format / Synthetic teaching material

Acute Extensive Anterior ST-Segment Elevation Myocardial Infarction

Successful Emergency Percutaneous Coronary Intervention and Drug-Eluting Stent Implantation

Medical Record Information

Item	Medical Record Information
Patient	Zhang Jianguo (fictional), male, 58 years old
Medical record number	SIM-AMI-20260901
Hospitalization period	September 1, 2026, 09:20 to September 7, 2026, 10:00
Key event	Acute thrombotic occlusion of the proximal left anterior descending artery; emergency implantation of 1 drug-eluting stent
Discharge outcome	Improved; LVEF improved from 45% to 50%
Document version	Expert-review edition v1.0 / Compilation date: 2026-09-10

Instructions for Use

This document follows the reading order of photocopied hospital medical records and the layout of laboratory reports. All data are synthetic and intended for teaching. It is solely for clinical experts to review the case's internal logic, the appropriateness of diagnosis and treatment, and its suitability for teaching. No real institutional identifiers, official seals, signatures, or actual patient information have been used.

Laboratory reports include specimen type, collection/receipt/report times, reference intervals, and arrows indicating abnormal results. ↑ indicates a value above the upper limit, and ↓ indicates a value below the lower limit; disease-specific treatment targets and laboratory reference intervals are explained separately.

The electrocardiograms are calibrated schematic waveforms drawn according to the case parameters, not device-acquired recordings. Echocardiography, chest radiography, and coronary angiography are presented as written reports with corresponding structural/motion diagrams; all images are explicitly labeled as synthetic.

Signature fields serve only as simulated role identifiers. This compilation is not a complete archived medical record with legal validity and does not include actual signed informed consent forms, payment documentation, or original records of nursing care performed.

Compiled by: Clinical Reasoning Training System. Clinical expert review: Not yet completed (both are role identifiers; no actual signatures have been made).

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SOURCE SECTION 2

Simulated Teaching Hospital

Registration Information and Data Synchronization Notes

Medical record copy format / Synthetic teaching material

Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Synchronization date: 2026-09-14. Source task: 1.2 Chest Pain Case-Overall Planning. The medical master draft remains v1.0, pending expert review; this synchronization is not case v1.2 and does not indicate that clinical review has been passed.

Registration details

Registration item	Current teaching scenario
Name / Sex / Age	Zhang Jianguo (fictional) / Male / 58 years
Ethnicity / Place of origin	Han Chinese / Hefei, Anhui Province
Marital status / Occupation	Married / Corporate administrative staff member
Current address	Room 301, Building 2, Chunhe Garden, 18 Kangning Road, Wuhou District, Chengdu, Sichuan Province (fictional for teaching)
History provided by	The patient himself
Registration status	Admission registration completed; registration information complete

The original medical course, surgery, actual medication use, and blood values have not been changed. The latest 38-item set of stage-specific reports, 68-item HIS medication catalog, and associated reference materials are available on the complete materials page in the case-generation section. The medication database version is 2026-09-11.2, which differs from the medical master draft version; inclusion of a medication does not indicate that the patient used it. The entire medication database remains pending local pharmacy verification.

Only fictional personal background details explicitly fixed in the task are retained; model improvisations, learner dialogues, examination procedures, or conversations about medical orders are not incorporated into the standard case. The original PDF is archived separately.

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SOURCE SECTION 3

Simulated Teaching Hospital

Admission Record (I)

Medical record copy format / Synthetic teaching material

Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

General Information

Zhang Jianguo, male, 58 years old, married, corporate administrative employee. Native place: Hefei, Anhui Province; currently resides in Chengdu. Admitted through the emergency department. History provided by the patient and considered reliable. Recorded at 2026-09-01 09:35; assessment and emergency treatment proceeded concurrently.

Chief Complaint

Persistent crushing retrosternal pain with profuse sweating for 1 hour and 20 minutes.

History of Present Illness

At 08:00 on 2026-09-01, while moving items, the patient suddenly developed crushing retrosternal pain over an area approximately the size of a palm, radiating to the left shoulder and medial aspect of the left upper arm. Pain severity was 8/10 points, accompanied by profuse sweating, nausea, and chest tightness. He stopped activity and rested for approximately 20 minutes without significant relief; the pain persisted. At 09:20, he arrived at this hospital's emergency department accompanied by family members, having received no prehospital medical treatment. At 09:25, an ECG showed ST-segment elevation in leads V1–V6, and the catheterization laboratory was immediately activated.

Pertinent Positive and Negative History

During the preceding 2 weeks, he had experienced 3 episodes of dull retrosternal pain while climbing stairs or walking briskly, each lasting 3–5 minutes and relieved by rest; he did not seek medical attention. During the current episode, there was no tearing chest or back pain, no pain related to breathing or posture, no fever, cough, hemoptysis, syncope, or impaired limb movement, and no unilateral lower-limb swelling or pain. No history of surgery, prolonged bed rest, or prolonged immobility during long-distance travel in the preceding 3 months. Before symptom onset, appetite and sleep were satisfactory, bowel movements and urination were normal, and there had been no significant recent weight change.

Past Medical History

Hypertension for 8 years, with a maximum blood pressure of approximately 170/100 mmHg. He intermittently takes amlodipine 5 mg once daily, with poor adherence; usual blood pressure is 145–160/90–100 mmHg. Elevated blood lipids were identified 2 years ago, without regular treatment. He denies a history of myocardial infarction, PCI, heart failure, diabetes mellitus, stroke, or chronic kidney disease. He denies a history of active ulcers, gastrointestinal bleeding, intracranial hemorrhage, coagulation disorders, or recent major trauma or surgery. No long-term anticoagulant use. He denies a history of asthma, severe chronic lung disease, hepatitis, tuberculosis, or blood transfusion.

Allergies and Recent Medication Use

He denies known allergies to medications or contrast agents. No recent use of PDE-5 inhibitors such as sildenafil.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures were made).

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Simulated Teaching Hospital

Admission Record (II)

Medical record photocopy format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Personal, Marital, Reproductive, and Family History

Has smoked for 30 years, approximately 20 cigarettes/day, totaling approximately 30 pack-years. Drinks alcohol occasionally, with no history of heavy drinking. Exercises little and consumes a salty, high-fat diet. No history of specific occupational exposures. Married with one son; spouse and child are healthy. His father had a myocardial infarction at age 54, and his mother has hypertension. Denies any known history of hereditary cardiomyopathy or family history of sudden death.

Vital Signs

Temperature 36.5°C, pulse 92 beats/min, respiratory rate 22 breaths/min; blood pressure 148/92 mmHg in the right arm and 146/90 mmHg in the left arm; SpO_2 96% on room air. Height 172 cm, weight 78 kg, BMI 26.4 kg/m^2.

Physical Examination

Alert, with an expression of acute distress and cool, clammy skin. No cyanosis of the lips or scleral icterus. No jugular venous distension; trachea midline. Breath sounds clear bilaterally, with no wheezes or crackles. Heart rate 92 beats/min, with a regular rhythm and slightly diminished heart sounds; no pathological murmurs, third heart sound, or pericardial friction rub. Abdomen flat and soft, without tenderness or rebound tenderness; liver and spleen not palpable below the costal margins. No edema or asymmetric swelling of the lower extremities. Radial and dorsalis pedis pulses symmetric bilaterally. No focal neurological abnormalities.

Initial Emergency Department Investigations

09:25 ECG: Sinus rhythm, ST-segment elevation in V1–V6, with reciprocal depression in III and aVF. Blood samples were collected at 09:28. Emergency reperfusion was initiated before the troponin result was available, without waiting for laboratory results.

Preliminary Admission Diagnoses

1. Coronary atherosclerotic heart disease: acute extensive anterior ST-segment elevation myocardial infarction, Killip class I. 2. Essential hypertension, grade 2, very high risk. 3. Dyslipidemia.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

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Simulation Teaching Hospital

Initial Progress Note

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Patient and Administrative Information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Recorded at	2026-09-01 09:40

Case Characteristics

A 58-year-old man with hypertension, dyslipidemia, long-term smoking, and a family history of premature coronary artery disease. He has had exertional chest pain over the past 2 weeks; the current episode of persistent chest pain has lasted more than 1 hour, accompanied by profuse sweating and nausea. There is marked ST-segment elevation in contiguous precordial leads, with reciprocal depression in the inferior leads. No pulmonary crackles, third heart sound, or hypotension.

Diagnosis and Supporting Evidence

Acute extensive anterior STEMI. Typical ischemic chest pain and localizing ST-segment elevation are sufficient to activate the emergency reperfusion pathway. Subsequent confirmation of the infarction mechanism requires serial troponin changes and angiography. Currently Killip class I.

Differential Diagnosis

1. Aortic dissection: Unlikely, given the absence of tearing pain, a marked interarm blood pressure difference, pulse deficits, and neurological symptoms. These negative findings alone cannot completely exclude it; promptly evaluate further if new clues emerge. 2. Pulmonary embolism: No major venous thrombosis risk factors, hypoxemia, or signs of right heart strain; the ECG is more consistent with coronary occlusion affecting the anterior wall. 3. Acute pericarditis: No positional pain or friction rub; ST changes are regional and reciprocal. 4. Myocarditis and stress cardiomyopathy: Assessment requires coronary and cardiac function investigations, but this must not delay current reperfusion. 5. Gastrointestinal disease cannot explain the localizing ECG changes.

Diagnostic and Treatment Plan

Proceed immediately with emergency coronary angiography and primary PCI as appropriate. Administer loading doses of aspirin and ticagrelor, heparin anticoagulation during the intervention, and intensive lipid-lowering therapy. Provide continuous ECG monitoring, establish intravenous access, and have defibrillation equipment ready. Monitor pain, blood pressure, oxygen saturation, bleeding, and signs of heart failure. Complete cardiac biomarker testing, a complete blood count, coagulation studies, biochemistry testing, and echocardiography; do not delay vessel reopening while awaiting results.

Risk Assessment

There is a risk of extensive anterior myocardial injury; remain alert for malignant arrhythmias, cardiogenic shock, mechanical complications, and death. There is no clear active bleeding, need for long-term anticoagulation, or other evident high-bleeding-risk feature. The antithrombotic regimen must be reassessed dynamically during hospitalization and follow-up.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B. Both are role identifiers; no actual signatures have been provided.

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Simulation Teaching Hospital

Emergency Resuscitation Record

Medical Record Copy Format / Synthetic Teaching Material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Resuscitation period: 2026-09-01 09:20 to 11:05. Resuscitation diagnosis: Acute extensive anterior-wall STEMI.

Clinical status and interventions

Time	Clinical status and interventions
09:20	Chest pain 8/10 points and profuse sweating; cardiac monitoring initiated and intravenous access established.
09:25 / 09:27	First ECG completed and STEMI identified; catheterization laboratory activated.
09:28	Samples collected for complete blood count, troponin, coagulation studies, liver and renal function, electrolytes, and lipid profile.
09:30	Aspirin, regular tablets, 300 mg chewed; ticagrelor 180 mg orally.
09:32	After checking blood pressure and contraindications, nitroglycerin 0.5 mg administered sublingually once; pain decreased to 6/10 points, BP 136/84 mmHg.
09:35 / 09:40	Atorvastatin 40 mg orally; PCI plan and risks explained, and simulated consent process completed.
09:50 / 09:52	hs-cTnI result reported: 186 ng/L; entered the catheterization laboratory.
10:04 / 10:12	Angiography showed complete occlusion of the proximal left anterior descending artery; guidewire crossed the lesion.
10:16 / 10:24	First balloon inflation restored antegrade flow; one drug-eluting stent implanted.
10:38 / 10:45	Final angiography showed TIMI grade 3 flow, and chest pain had essentially resolved; procedure completed.
11:05	Transferred to CCU, with stable circulation.

Arrival-to-ECG time: 5 minutes; arrival-to-guidewire crossing time: 52 minutes; arrival-to-first balloon inflation time: 56 minutes; symptom onset-to-first balloon inflation time: 2 hours 16 minutes.

SpO2 remained 95%–97% throughout; supplemental oxygen was not routinely administered. There was no cardiac arrest, sustained ventricular tachycardia/ventricular fibrillation, or shock; no chest compressions, defibrillation, or vasopressor therapy were performed or administered. Successful resuscitation refers to successful emergency reperfusion, symptom relief, and stable circulation.

Resuscitation physician: Simulated Physician A. Assisting physician: Simulated Physician B (both are role identifiers; no actual signatures were made).

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Simulation Teaching Hospital

Preoperative Record and Summary of Informed Consent Discussion

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Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Record time	2026-09-01 09:45
Planned procedure time	Emergency procedure on the same day		

Preoperative Diagnosis

Acute extensive anterior ST-segment elevation myocardial infarction.

Indications for the Procedure

Symptoms for less than 12 hours, ongoing ischemic chest pain, definite ST-segment elevation, and availability of immediate primary PCI.

Planned Procedure

Selective coronary angiography, with coronary balloon angioplasty and stent implantation if necessary, and intravascular imaging as needed.

Preoperative Assessment

The patient is alert and able to express himself independently. BP is stable, with no evidence of pulmonary congestion or shock. Loading doses of dual antiplatelet therapy have been administered. There is no known contrast agent allergy, active bleeding, or history of recent major surgery. Laboratory testing and investigations were performed concurrently with emergency preparations; reperfusion was not delayed while awaiting renal function or other results.

Content of the Discussion

The life-threatening nature of acute anterior myocardial infarction, the benefits of emergency vessel reopening, and alternative strategies including medical therapy and thrombolysis were explained. Timely PCI is available at this hospital, and primary PCI was selected. Risks including bleeding, vascular injury, allergic reactions, kidney injury, coronary dissection/perforation, no-reflow, stent thrombosis, emergency surgery, and death were explained. The patient indicated understanding and consented; family members participated in the discussion.

Anesthesia and Preparation

Local anesthesia, with right radial artery access preferred; verify identity, planned procedure, allergy history, antiplatelet loading, and resuscitation equipment.

Document Status

This page is a simulated summary of the consent discussion and does not replace an informed consent form for an interventional procedure signed by a real patient; no real signatures are generated or imitated.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

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Simulated Teaching Hospital

Coronary Intervention Procedure Record (I)

Medical record copy format / Synthetic teaching material

Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Procedure details

Field	Value
Date	2026-09-01
Start time	09:55
End time	10:45
Anesthesia	Local anesthesia
Procedure	Coronary angiography + left anterior descending artery balloon angioplasty + drug-eluting stent implantation + intravascular ultrasound (IVUS).

Access, Anticoagulation, and Angiography

After routine skin preparation and draping, local infiltration anesthesia was administered with 1% lidocaine, and a 6 F sheath was inserted via the right radial artery. Unfractionated heparin 7,000 U was administered intravenously, approximately 90 U/kg; intraoperative ACT was 286 seconds. No glycoprotein IIb/IIIa antagonist was used.

Coronary angiography findings

Vessel	Angiographic findings
Coronary dominance	Right-dominant
Left main coronary artery (LM)	No significant stenosis.
Left anterior descending artery (LAD)	Total occlusion of the proximal segment before the origin of the first diagonal branch, with an irregular stump and signs of thrombus; TIMI grade 0.
Left circumflex artery (LCX)	Approximately 30% stenosis in the mid segment; TIMI grade 3.
Right coronary artery (RCA)	Approximately 40% stenosis in the mid segment; TIMI grade 3.

Lesion Treatment

The left anterior descending artery was identified as the infarct-related artery. A working guidewire was advanced through a 6 F EBU 3.5 guiding catheter and crossed the occlusion into the distal left anterior descending artery at 10:12. At 10:16, predilation with a 2.0 × 15 mm balloon restored TIMI grade 2 flow, revealing approximately 95% residual stenosis and a lesion length of approximately 24 mm.

IVUS was performed after restoration of blood flow: the lesion segment showed a substantial plaque burden with signs of intraluminal thrombus; the distal reference diameter was approximately 3.0 mm and the proximal reference diameter approximately 3.5 mm. Angiography and IVUS are not equivalent to histopathological examination; plaque rupture was not documented as confirmed.

Operator: Simulated Physician A. Assistant: Simulated Physician B (both are role identifiers; no actual signatures were provided).

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Simulated Teaching Hospital

Coronary Intervention Procedure Record (II)

Medical record copy format / Synthetic teaching material

Name: Zhang Jianguo (fictional) Sex: Male Age: 58 years Medical record number: SIM-AMI-20260901

Department: Cardiology / CCU Admission: 2026-09-01 09:20 Discharge: 2026-09-07 10:00

Stent Implantation and Results

At 10:24, one 3.0 × 28 mm drug-eluting stent was implanted and deployed at 12 atm. The proximal and middle segments were post-dilated with a 3.5 × 12 mm noncompliant balloon at a maximum pressure of 16 atm, avoiding extension beyond the stent edges.

Repeat IVUS: satisfactory stent expansion and apposition, with a minimum stent area of approximately 7.1 mm^2 and no significant edge dissection. Final angiography at 10:38: residual stenosis <10%, TIMI grade 3 flow, patent diagonal branch, and no perforation, dissection, or significant distal embolization. The nonculprit vessels had no severe stenosis and did not require intervention during this procedure. Routine thrombus aspiration was not performed.

Procedure Details and Results

Item	Result
Stent	One drug-eluting stent, 3.0 × 28 mm; no manufacturer, lot number, or UDI is fabricated.
Contrast agent	Iodixanol 110 mL
Intraprocedural ACT	286 seconds (a monitoring value during procedural anticoagulation; reference intervals for patients not receiving anticoagulation are not used to classify this value as abnormal)
Blood pressure	118–142/72–88 mmHg
Estimated blood loss	<20 mL
Symptoms	Chest pain decreased from 6/10 points to 0–1/10 points
Complications	No sustained malignant arrhythmia, perforation, no-reflow, or similar complications occurred
Hemostasis	The sheath was removed at the end of the procedure, and hemostasis was achieved using a radial artery compression device; distal circulation in the hand was good

Postprocedural Diagnosis and Management

Acute extensive anterior STEMI; acute thrombotic occlusion of the proximal left anterior descending artery; successful primary PCI. Transferred to the CCU at 11:05, with continued dual antiplatelet and lipid-lowering therapy. There was no other indication for anticoagulation; heparin was not routinely continued after the procedure. Monitor the puncture site, cardiac rhythm, ECG, hemoglobin, urine output, and renal function.

Operator: Simulated Physician A Assistant: Simulated Physician B (both are role identifiers; no actual signatures were provided)

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Simulated Teaching Hospital

Summary of Principal Medical Orders

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Name: Zhang Jianguo (fictional) Sex: Male Age: 58 years Medical record number: SIM-AMI-20260901

Department: Cardiology / CCU Admission: 2026-09-01 09:20 Discharge: 2026-09-07 10:00

Principal Medical Orders

Medication/Measure	Dose and Route	Start/Stop and Notes
Aspirin	300 mg loading dose, chewed; 100 mg orally once daily	Loading dose on September 1; maintenance from September 2
Ticagrelor	180 mg loading dose; 90 mg orally twice daily	Loading dose at 09:30 on September 1; maintenance from 21:30 that day, approximately every 12 hours
Unfractionated heparin	7,000 U intravenously	During the intervention only, with ACT monitoring; discontinued after the procedure
Atorvastatin	40 mg orally once nightly	First dose at 09:35 on September 1, not repeated that day; nightly from September 2
Ezetimibe	10 mg orally once daily	From September 2
Metoprolol extended-release tablets	23.75 mg once daily, subsequently 47.5 mg once daily	Low dose from September 2; dose increased on September 4
Perindopril tert-butylamine tablets	2 mg once daily, subsequently 4 mg once daily	From September 2; dose increased on September 4
Nitroglycerin	0.5 mg sublingually	Single dose at 09:32 on September 1; supplied for use as needed after discharge
Monitoring and Activity	Continuous ECG monitoring for the first 48 hours; staged activity after stabilization	Transferred to the general ward on September 3; inpatient phase I rehabilitation

This table summarizes the principal medical orders and is not an actual nursing medication administration record. Perindopril is explicitly specified as the tert-butylamine salt formulation to avoid confusion with other salt forms.

There was no clinical pulmonary congestion, and no diuretic was used. With LVEF >40% and no diabetes or clinical heart failure, an aldosterone receptor antagonist was not routinely added. There were no evident gastrointestinal bleeding risk factors such as active peptic ulcer, a history of bleeding, or concomitant anticoagulant/NSAID use; a PPI was not routinely added at this time, with reassessment planned.

Recording physician: Simulated Physician A Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made)

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Simulation Teaching Hospital

Progress Notes (1)

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Patient and Admission Details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

2026-09-01 11:30 First Postprocedural Progress Note

Chest pain, profuse sweating, and nausea have resolved. BP 126/78 mmHg, HR 78 beats/min, SpO2 97%; both lungs clear; heart rhythm regular. No bleeding at the right radial artery puncture site; fingers warm with normal sensation. TIMI grade 3 flow after stenting of the left anterior descending artery; ECG at 11:20 showed marked resolution of ST elevation. Continue dual antiplatelet therapy, statin therapy, and monitoring; repeat serial biomarker measurements; remain alert for recurrent ischemia, heart failure, arrhythmias, and bleeding; complete bedside echocardiography.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

2026-09-01 18:00 Postprocedural Observation

No chest pain or shortness of breath. BP 122/76 mmHg, HR 74 beats/min; urine output normal. No oozing at the puncture site; compression reduced according to protocol; distal perfusion good. Echocardiography showed LVEF 45%, with hypokinesis of the anterior wall, anterior septum, and apex, and no mechanical complications. Occasional isolated premature ventricular contractions, with no ventricular couplets or sustained ventricular tachycardia; no prophylactic antiarrhythmic medication is to be used.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

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Simulated Teaching Hospital

Progress Notes (2)

Medical record photocopy format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

2026-09-02 09:00 Senior Physician Ward Round

No chest pain or dyspnea overnight. BP 124/78 mmHg, HR 72 beats/min; no crackles in either lung. hs-cTnI 32,400 ng/L, the highest sampled value; Q waves and T-wave inversion on ECG are consistent with the evolution of the infarction. With no new symptoms or recurrent ST elevation, reinfarction is not currently suspected. Type 1 STEMI is confirmed in conjunction with the coronary findings. Despite mildly reduced LVEF, there is no clinical heart failure; Killip class I. Add extended-release metoprolol tablets 23.75 mg once daily and perindopril tert-butylamine tablets 2 mg once daily; monitor heart rate, blood pressure, creatinine, and potassium. LDL-C 4.05 mmol/L; add ezetimibe 10 mg once daily.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

2026-09-03 09:00 Daily Progress Note and Transfer Out of CCU

No chest pain, palpitations, or shortness of breath; able to sit up and walk short distances. BP 120/74 mmHg, HR 68 beats/min; both lungs clear. Slight bruising at the puncture site, with no hematoma. Biomarkers are declining; creatinine 84 μmol/L, hemoglobin 143 g/L; no AKI or clinical bleeding. No sustained malignant arrhythmias during 48 hours of monitoring. Transfer to the general ward, with handover of dual antiplatelet therapy, bleeding monitoring, and the rehabilitation plan.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

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Simulated Teaching Hospital

Progress Notes (3)

Medical record copy format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

2026-09-04 09:00 Senior Physician Ward Round

No chest pain or shortness of breath; tolerates walking in the ward. BP 128/78 mmHg, HR 74 beats/min; no significant dizziness on standing. CK-MB has returned to the reference range. Metoprolol increased to 47.5 mg once daily, and perindopril tert-butylamine tablets increased to 4 mg once daily. Monitor for hypotension, bradycardia, hyperkalemia, and changes in renal function. No wheezing, atrioventricular block, or other drug intolerance.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

2026-09-05 09:00 Daily Progress Note

No chest pain, palpitations, or melena. BP 118/72 mmHg, HR 66 beats/min. Walks in separate sessions of 5–10 minutes each without discomfort. Medication names, doses, and timing have been verified with the patient and family; emphasized that antiplatelet medication must not be discontinued without medical advice. The patient agrees to stop smoking. No clinical manifestations of infection or venous thrombosis. Continue rehabilitation; repeat ECG, echocardiography, and blood tests are planned before discharge.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

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Simulation Teaching Hospital

Clinical Progress Notes (4)

Medical record reproduction format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

2026-09-06 09:00 Pre-discharge Assessment

No chest pain, shortness of breath, or syncope. BP 116/72 mmHg, HR 64 beats/min. Hb 145 g/L, PLT 229×10^{9}/L, Cr 80 μmol/L, K 4.3 mmol/L; liver function has recovered. hs-cTnI 780 ng/L, still above the upper limit but continuing to decline, with no evidence of recurrent ischemia. Echocardiography showed LVEF 50%, with no definite left ventricular thrombus seen. Meets discharge criteria; rehabilitation and follow-up have been arranged.

Documenting physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been provided).

2026-09-07 09:00 Discharge-day Progress Note

No chest pain, palpitations, or shortness of breath. BP 118/74 mmHg, HR 66 beats/min, SpO_2 97% on room air. Both lungs clear, heart rhythm regular, and no lower-extremity edema. The puncture site is healing well, and the patient is independent in daily activities. Discharge medications and the follow-up plan were verified. The patient can repeat back how to respond to recurrent chest pain and warning symptoms of bleeding. Discharge today at 10:00.

Documenting physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been provided).

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Simulated Teaching Hospital

Discharge Record (I)

Medical record copy format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Condition on Admission

Admitted with persistent crushing retrosternal pain accompanied by profuse sweating for 1 hour 20 minutes. History of hypertension, dyslipidemia, and long-term smoking. ECG showed ST-segment elevation in V1–V6; initial hs-cTnI was 186 ng/L. No pulmonary congestion or shock.

Hospital Course

The STEMI pathway was activated, and loading doses of dual antiplatelet therapy and lipid-lowering treatment were administered. Emergency angiography showed proximal left anterior descending artery occlusion with TIMI grade 0 flow. One 3.0 × 28 mm drug-eluting stent was implanted, achieving postprocedural TIMI grade 3 flow. Chest pain resolved, and ST-segment elevation decreased markedly. Cardiac biomarkers initially rose and then fell. No sustained malignant arrhythmias, clinical heart failure, mechanical complications, major bleeding, or AKI occurred. Dual antiplatelet therapy, intensive combination lipid-lowering therapy, a β-blocker, and an ACE inhibitor were continued. Early rehabilitation and secondary prevention education were completed. LVEF improved from 45% to 50%.

Discharge Diagnoses

1. Coronary atherosclerotic heart disease: acute extensive anterior-wall STEMI, type 1; acute thrombotic occlusion of the proximal left anterior descending artery; status post emergency PCI and drug-eluting stent implantation; Killip class I on admission. 2. Essential hypertension, grade 2, very high risk. 3. Dyslipidemia.

Condition at Discharge

Alert, with no chest pain, shortness of breath, or palpitations. BP 118/74 mmHg; HR 66 beats/min. Both lungs clear, no lower-extremity edema, and the radial artery puncture site was healing well. Able to tolerate light daily physical activity. Outcome: improved.

Basis for Discharge

Revascularization was successful, with no ongoing ischemia or complications. Medical therapy was tolerated, and rehabilitation and follow-up arrangements were in place. Troponin had not yet normalized, and Q waves and regional wall-motion abnormalities persisted; these do not preclude discharge when clinically stable. Stent implantation does not indicate complete myocardial recovery or a cure for coronary artery disease.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been provided).

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Simulated Teaching Hospital

Discharge Record (II): Medications and Follow-up

Medical record photocopy format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Discharge medications

Medication	Discharge instructions	Reassessment notes
Enteric-coated aspirin tablets	100 mg once daily; swallow whole	Currently planned for long-term use; adjust according to the antithrombotic strategy
Ticagrelor	90 mg twice daily, approximately every 12 hours	The default plan is to combine with aspirin for at least 12 months; reassess around September 2027, with ongoing assessment of bleeding in the meantime
Atorvastatin	40 mg once every evening	Long-term use; adjust according to lipid levels and tolerability
Ezetimibe	10 mg once daily	Combination lipid-lowering therapy; adjust after repeat testing
Extended-release metoprolol tablets	47.5 mg once daily	Reassess according to heart rate, blood pressure, LVEF, and symptoms
Perindopril tert-butylamine tablets	4 mg once daily	Adjust according to blood pressure, potassium, and renal function
Nitroglycerin	0.5 mg sublingually, as needed	Use as instructed; not a substitute for calling 120

All routine medications above are taken orally; nitroglycerin is administered sublingually. A lifelong fixed dose of metoprolol is not presumed for all patients with myocardial infarction. Do not stop antiplatelet therapy on your own; surgery, tooth extraction, and bleeding require prior coordination with cardiology. Avoid self-medication with NSAIDs such as ibuprofen and diclofenac.

Follow-up plan

Cardiology follow-up in 7–14 days: assess symptoms, heart rate, blood pressure, adherence, and bleeding; recheck creatinine and potassium, and obtain a complete blood count if necessary. Recheck lipids in 4–6 weeks and liver function as indicated; the LDL-C target is <1.4 mmol/L and a reduction of ≥50% from baseline. If the target is not reached, assess adherence and intensify treatment. Repeat echocardiography in 6–12 weeks.

Refer for outpatient cardiac rehabilitation. Stop smoking, avoid secondhand smoke, limit salt and saturated fat, and gradually work toward weight control. Begin with walking at a comfortable pace; avoid sudden strenuous exercise, heavy lifting, and breath-holding. Record blood pressure and heart rate; the usual blood pressure target is <130/80 mmHg, if tolerated.

If similar chest pain recurs, especially if it lasts >5 minutes or is accompanied by profuse sweating, shortness of breath, or syncope: call 120 immediately; do not drive yourself. Use the as-needed nitroglycerin as instructed if there is no hypotension or relevant contraindication; do not wait until after repeated doses to call for help. Seek emergency care immediately for vomiting blood, black stools, persistent bleeding, or sudden severe headache or hemiplegia, and inform the emergency staff about antiplatelet use.

Recording physician: Simulated Physician A. Reviewing physician: Simulated Physician B (both are role identifiers; no actual signatures have been made).

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SOURCE SECTION 17

Simulation Teaching Hospital

Hematology Laboratory Report

Medical record copy format / Synthetic teaching material

Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Report and specimen information

Field	Value
Report number	SIM-CBC-0901
Specimen	EDTA-anticoagulated whole blood
Specimen status	Acceptable (simulated)
Collection	09-01 09:28
Received	09:32
Reported	09:45

Hematology test results

Test	Result	Flag	Unit	Reference interval
White blood cell count WBC	12.6	↑	$\times10^9$/L	3.5 - 9.5
Neutrophil percentage	82	↑	%	40 - 75
Lymphocyte percentage	11	↓	%	20 - 50
Monocyte percentage	6		%	3 - 10
Eosinophil percentage	0.5		%	0.4 - 8
Basophil percentage	0.5		%	0 - 1
Neutrophil count	10.33	↑	$\times10^9$/L	1.8 - 6.3
Lymphocyte count	1.39		$\times10^9$/L	1.1 - 3.2
Monocyte count	0.76	↑	$\times10^9$/L	0.1 - 0.6
Eosinophil count	0.06		$\times10^9$/L	0.02 - 0.52
Basophil count	0.06		$\times10^9$/L	0 - 0.06
Red blood cell count RBC	4.9		$\times10^{12}$/L	4.3 - 5.8
Hemoglobin Hb	148		g/L	130 - 175
Hematocrit HCT	44.1		%	40 - 50
Mean corpuscular volume MCV	90.0		fL	82 - 100
Mean corpuscular hemoglobin MCH	30.2		pg	27 - 34
Mean corpuscular hemoglobin concentration MCHC	336		g/L	316 - 354
Red cell distribution width RDW-CV	12.8		%	11.5 - 14.5
Platelet count PLT	236		$\times10^9$/L	125 - 350
Mean platelet volume MPV	10.2		fL	7 - 13

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

The newly added differential counts and red blood cell indices are supplementary synthetic data for this edition, consistent with the existing WBC/Hb/PLT values.

Reference intervals are set by this simulated laboratory and apply only to this teaching case; they are not from the instructions for any specific brand of reagent.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures have been provided).

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SOURCE SECTION 18

Simulation Teaching Hospital

Hematology Laboratory Report

Medical record copy format / Synthetic teaching material

Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Report and specimen information

Field	Value
Report number	SIM-CBC-0903
Specimen	EDTA-anticoagulated whole blood
Specimen status	Acceptable (simulated)
Collection	09-03 06:00
Received	06:12
Reported	06:40

Hematology test results

Test	Result	Flag	Unit	Reference interval
White blood cell count (WBC)	9.4		$\times10^9$/L	3.5 - 9.5
Neutrophil percentage	71		%	40 - 75
Lymphocyte percentage	20		%	20 - 50
Monocyte percentage	7		%	3 - 10
Eosinophil percentage	1.5		%	0.4 - 8
Basophil percentage	0.5		%	0 - 1
Neutrophil count	6.67	↑	$\times10^9$/L	1.8 - 6.3
Lymphocyte count	1.88		$\times10^9$/L	1.1 - 3.2
Monocyte count	0.66	↑	$\times10^9$/L	0.1 - 0.6
Eosinophil count	0.14		$\times10^9$/L	0.02 - 0.52
Basophil count	0.05		$\times10^9$/L	0 - 0.06
Red blood cell count (RBC)	4.75		$\times10^{12}$/L	4.3 - 5.8
Hemoglobin (Hb)	143		g/L	130 - 175
Hematocrit (HCT)	42.8		%	40 - 50
Mean corpuscular volume (MCV)	90.1		fL	82 - 100
Mean corpuscular hemoglobin (MCH)	30.1		pg	27 - 34
Mean corpuscular hemoglobin concentration (MCHC)	334		g/L	316 - 354
Red cell distribution width (RDW-CV)	12.9		%	11.5 - 14.5
Platelet count (PLT)	221		$\times10^9$/L	125 - 350
Mean platelet volume (MPV)	10.1		fL	7 - 13

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

The newly added differential counts and red blood cell indices are supplemental synthetic data for this edition and are consistent with the existing WBC/Hb/PLT values.

Reference intervals were established by this simulated laboratory and apply only to this teaching case; they are not from any specific brand's reagent instructions for use.

Performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures have been provided).

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Simulation Teaching Hospital

Hematology Laboratory Report

Medical record copy format / Synthetic teaching material

Patient and report details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Report number	SIM-CBC-0906
Specimen	EDTA-anticoagulated whole blood	Specimen status	Acceptable (simulated)
Collection	09-06 06:00	Received	06:12
Reported	06:40		

Hematology results

Test	Result	Flag	Unit	Reference interval
White blood cell count (WBC)	7.2		$\times10^9$/L	3.5 - 9.5
Neutrophil percentage	63		%	40 - 75
Lymphocyte percentage	27		%	20 - 50
Monocyte percentage	7		%	3 - 10
Eosinophil percentage	2.5		%	0.4 - 8
Basophil percentage	0.5		%	0 - 1
Neutrophil count	4.54		$\times10^9$/L	1.8 - 6.3
Lymphocyte count	1.94		$\times10^9$/L	1.1 - 3.2
Monocyte count	0.5		$\times10^9$/L	0.1 - 0.6
Eosinophil count	0.18		$\times10^9$/L	0.02 - 0.52
Basophil count	0.04		$\times10^9$/L	0 - 0.06
Red blood cell count (RBC)	4.81		$\times10^{12}$/L	4.3 - 5.8
Hemoglobin (Hb)	145		g/L	130 - 175
Hematocrit (HCT)	43.5		%	40 - 50
Mean corpuscular volume (MCV)	90.4		fL	82 - 100
Mean corpuscular hemoglobin (MCH)	30.1		pg	27 - 34
Mean corpuscular hemoglobin concentration (MCHC)	333		g/L	316 - 354
Red cell distribution width (RDW-CV)	12.8		%	11.5 - 14.5
Platelet count (PLT)	229		$\times10^9$/L	125 - 350
Mean platelet volume (MPV)	10.2		fL	7 - 13

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

The newly added differential counts and red cell indices are supplementary synthetic data for this edition and are consistent with the existing WBC/Hb/PLT values.

Reference intervals were established by this simulated laboratory and apply only to this teaching case; they are not taken from the instructions for any specific brand of reagent.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures have been provided).

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Simulated Teaching Hospital

Biochemistry Laboratory Report

Medical record copy format / Synthetic teaching material

Patient and specimen information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Report number	SIM-BIO-0901
Specimen	Serum	Specimen status	Acceptable (simulated)
Collection	09-01 09:28	Received	09:32
Reported	10:00		

Biochemistry results

Test	Result	Flag	Unit	Reference interval
Potassium K	3.8		mmol/L	3.5 - 5.3
Sodium Na	139		mmol/L	137 - 147
Magnesium Mg	0.84		mmol/L	0.75 - 1.02
Creatinine Cr	82		μmol/L	57 - 97
Estimated glomerular filtration rate	95		mL/min/1.73 m^2	≥90
Urea	5.6		mmol/L	3.1 - 8.0
Alanine aminotransferase ALT	28		U/L	9 - 50
Aspartate aminotransferase AST	36		U/L	15 - 40
Total bilirubin TBil	12.6		μmol/L	5 - 21
Albumin ALB	43		g/L	40 - 55

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

eGFR is estimated using CKD-EPI and rounded; a single result cannot be used to diagnose chronic kidney disease. Albumin was not retested on September 3.

Reference intervals are set by this simulated laboratory and apply only to this teaching case; they are not from the instructions for any specific brand of reagent.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were made).

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Simulated Teaching Hospital

Biochemistry Laboratory Report

Medical record reproduction format / Synthetic teaching material

Patient and report details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Report number	SIM-BIO-0903
Specimen	Serum	Specimen status	Acceptable (simulated)
Collection	09-03 06:00	Receipt	06:12
Reported	07:05		

Biochemistry results

Test	Result	Flag	Unit	Reference interval
Potassium K	4.2		mmol/L	3.5 - 5.3
Sodium Na	140		mmol/L	137 - 147
Magnesium Mg	0.89		mmol/L	0.75 - 1.02
Creatinine Cr	84		μmol/L	57 - 97
Estimated glomerular filtration rate	92		mL/min/1.73m^2	≥90
Urea	5.4		mmol/L	3.1 - 8.0
Alanine aminotransferase ALT	31		U/L	9 - 50
Aspartate aminotransferase AST	78	↑	U/L	15 - 40
Total bilirubin TBil	11.8		μmol/L	5 - 21

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

eGFR is estimated using CKD-EPI and rounded; a single result cannot be used to diagnose chronic kidney disease. Albumin was not retested on September 3.

Reference intervals are set by this simulated laboratory and apply only to this teaching case; they are not taken from any specific brand's reagent instructions.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were provided).

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Simulated Teaching Hospital

Biochemistry Laboratory Report

Medical record copy format / Synthetic teaching material

Patient and specimen information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Report number	SIM-BIO-0906
Specimen	Serum	Specimen status	Acceptable (simulated)
Collection	09-06 06:00	Received	06:12
Reported	07:05		

Biochemistry results

Test	Result	Flag	Unit	Reference interval
Potassium K	4.3		mmol/L	3.5 - 5.3
Sodium Na	140		mmol/L	137 - 147
Magnesium Mg	0.9		mmol/L	0.75 - 1.02
Creatinine Cr	80		μmol/L	57 - 97
Estimated glomerular filtration rate	98		mL/min/1.73m^2	≥90
Urea	5.2		mmol/L	3.1 - 8.0
Alanine aminotransferase ALT	30		U/L	9 - 50
Aspartate aminotransferase AST	29		U/L	15 - 40
Total bilirubin TBil	12.1		μmol/L	5 - 21
Albumin ALB	41		g/L	40 - 55

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

eGFR is estimated using CKD-EPI and rounded; a single result cannot be used to diagnose chronic kidney disease. Albumin was not retested on September 3.

Reference intervals are set by this simulated laboratory; they apply only to this teaching case and are not taken from any specific brand's reagent instructions for use.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were provided).

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Simulated Teaching Hospital

Lipid Panel Report

Medical record copy format / Synthetic teaching material

Patient and report information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Report number	SIM-LIP-0901
Specimen	Serum (on admission, before lipid-lowering treatment)	Specimen status	Acceptable (simulated)
Collection	09-01 09:28	Received	09:32
Reported	10:10		

Lipid test results

Test	Result	Flag	Unit	Reference interval
Total cholesterol (TC)	6.12	↑	mmol/L	<5.2 (desirable level)
Low-density lipoprotein cholesterol (LDL-C)	4.05	↑	mmol/L	<3.4 (general population)
High-density lipoprotein cholesterol (HDL-C)	0.98	↓	mmol/L	≥1.0
Triglycerides (TG)	1.86	↑	mmol/L	<1.7
Non-HDL-C	5.14	↑	mmol/L	<4.1 (general population)

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

This case involves secondary prevention of ACS: the LDL-C treatment goal is <1.4 mmol/L and a reduction of ≥50% from baseline, which is stricter than the thresholds indicated for the general population in the table. The sample was collected in a nonfasting state; TG should be interpreted in light of the sampling status.

The reference intervals were established by this simulated laboratory and are intended solely for this teaching case; they are not from the instructions for any specific brand of reagent.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were provided).

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Simulated Teaching Hospital

Coagulation Test Report

Medical record copy format / Synthetic teaching material

Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Report and specimen information

Field	Value
Report number	SIM-COAG-0901
Specimen	Sodium citrate-anticoagulated plasma
Specimen status	Acceptable (simulated)
Collection	09-01 09:28
Received	09:32
Reported	09:55

Coagulation test results

Test	Result	Flag	Unit	Reference interval
Prothrombin time (PT)	11.8		s	10 - 14
International normalized ratio (INR)	1.02			0.80 - 1.20
Activated partial thromboplastin time	29		s	25 - 35
Fibrinogen (FIB)	3.4		g/L	2.0 - 4.0

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is within the listed interval.

This specimen was collected before heparin administration. The intraoperative ACT of 286 seconds is documented in the procedure record and cannot be interpreted using coagulation reference intervals for patients not receiving anticoagulation.

The reference intervals were established by this simulated laboratory and apply only to this teaching case; they are not taken from the instructions for any specific brand of reagent.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were provided).

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Simulated Teaching Hospital

Compilation of Myocardial Injury Marker Reports

Medical record copy format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Specimen: serum; hs-cTnI method: high-sensitivity immunoassay (simulated); CK-MB: mass concentration assay.

SIM-MARK-01 - Collected: 09-01 09:28; Received: 09:32; Reported: 09:50

Test	Result	Flag	Upper reference limit
hs-cTnI	186 ng/L	↑	Male 99th percentile: 34 ng/L
CK-MB	8.6 ng/mL	↑	5.0 ng/mL

SIM-MARK-02 - Collected: 09-01 15:00; Received: 15:10; Reported: 15:40

Test	Result	Flag	Upper reference limit
hs-cTnI	18600 ng/L	↑	Male 99th percentile: 34 ng/L
CK-MB	126 ng/mL	↑	5.0 ng/mL

SIM-MARK-03 - Collected: 09-02 06:00; Received: 06:12; Reported: 06:45

Test	Result	Flag	Upper reference limit
hs-cTnI	32400 ng/L	↑	Male 99th percentile: 34 ng/L
CK-MB	184 ng/mL	↑	5.0 ng/mL

This page compiles the reports for the first three separate specimens. Upper reference limits are as defined by this simulated laboratory.

Tested by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were provided).

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Simulation Teaching Hospital

Compilation of Myocardial Injury Marker Reports (Continued)

Medical Record Copy Format / Synthetic Teaching Material

Patient and admission information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

Test specimen: serum; the following are reports for three separate specimens collected during the later part of the hospital stay.

SIM-MARK-04 Collected: 09-03 06:00 Received: 06:12 Reported: 06:45

SIM-MARK-04 test results

Test	Result	Flag	Upper reference limit
hs-cTnI	14800 ng/L	↑	Male 99th percentile: 34 ng/L
CK-MB	48 ng/mL	↑	5.0 ng/mL

SIM-MARK-05 Collected: 09-04 06:00 Received: 06:12 Reported: 06:45

SIM-MARK-05 test results

Test	Result	Flag	Upper reference limit
hs-cTnI	5260 ng/L	↑	Male 99th percentile: 34 ng/L
CK-MB	4.6 ng/mL		5.0 ng/mL

SIM-MARK-06 Collected: 09-06 06:00 Received: 06:12 Reported: 06:45

SIM-MARK-06 test results

Test	Result	Flag	Upper reference limit
hs-cTnI	780 ng/L	↑	Male 99th percentile: 34 ng/L
CK-MB	Not retested		5.0 ng/mL

The above is a compilation of results from 6 separate specimens, not a single test run; troponin does not have to return to normal before discharge. All specimen dates are in 2026.

Test performed by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were provided).

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Simulated Teaching Hospital

Compilation of Glucose Metabolism, Heart Failure Biomarker, and Urinalysis Reports

Medical record copy format / Synthetic teaching material

Patient and report information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Report number	SIM-OTHER
Specimen	See test descriptions	Specimen status	Acceptable (simulated)
Collection	See table below	Receipt	Approximately 10 minutes after collection
Reporting	See notes		

Laboratory results

Test	Result	Flag	Unit	Reference interval
Random blood glucose (09-01 09:28)	8.2	↑	mmol/L	3.9 - 7.8*
Fasting blood glucose (09-02 06:00)	5.7		mmol/L	3.9 - 6.1
Fasting blood glucose (09-06 06:00)	5.5		mmol/L	3.9 - 6.1
HbA1c (09-02 06:00)	5.6		%	4.0 - 6.0
NT-proBNP (09-01 09:28)	210	↑	pg/mL	<125*
NT-proBNP (09-02 06:00)	680	↑	pg/mL	<125*
NT-proBNP (09-06 06:00)	260	↑	pg/mL	<125*
Urine protein (09-02 06:20)	Negative			Negative
Urine occult blood (same as above)	Negative			Negative
Urine glucose (same as above)	Negative			Negative

Result flags: ↑ Above the upper reference limit; ↓ Below the lower reference limit; a blank indicates that the result is not outside the listed interval.

*The random blood glucose arrow is based solely on this simulated laboratory's flagging cutoff and cannot be used to diagnose diabetes. NT-proBNP uses a reference cutoff for the non-acute setting, not a diagnostic threshold for acute heart failure; this case has no clinical heart failure. Serum is used for blood glucose and NT-proBNP, EDTA whole blood for HbA1c, and morning urine for routine urinalysis. Each test is reported approximately 60 minutes after collection; this page is a compilation, not an original report for a single specimen.

Reference intervals are set by this simulated laboratory and apply only to this teaching case; they are not instructions for any specific brand of reagent.

Tested by: Simulated Laboratory Technologist A. Reviewer: Simulated Laboratory Technologist B (both are role identifiers; no actual signatures were made).

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Simulated Teaching Hospital

12-Lead Electrocardiogram Report

Medical record photocopy format / Synthetic teaching material

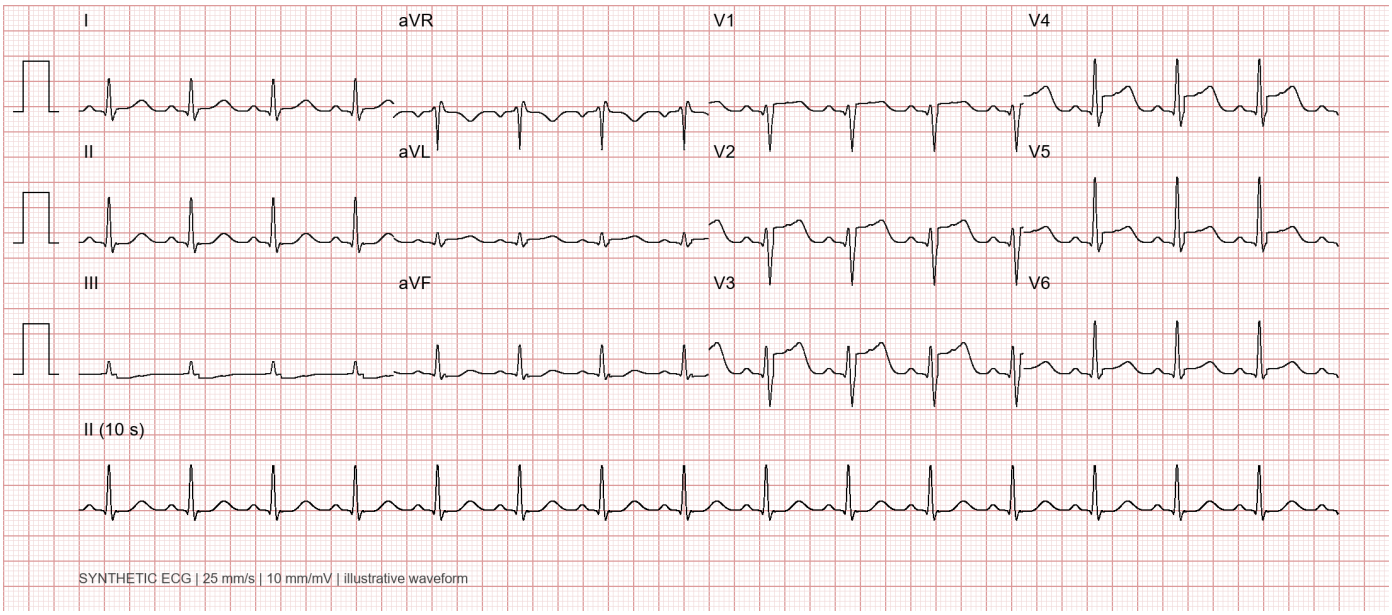
Patient and examination details

Name	Sex / Age	Case ID	Examination date and time	Heart rate	Phase
Zhang Jianguo (fictional)	Male / 58 years	SIM-AMI-20260901	2026-09-01 09:25	92 beats/min	Acute phase

Sinus rhythm; PR 160 ms, QRS 90 ms (case-specified values). ST elevation in V1–V6 is approximately 1.5, 3, 4, 3, 2, and 1 mm, respectively; mild elevation in I and aVL, and depression of approximately 0.5–1 mm in III and aVF; tall, peaked T waves in V2–V4. Impression: Acute extensive anterior wall STEMI.

Synthetic schematic waveforms, not acquired by a device; the quantitative values in this report are case-specified and must not be replaced by automated measurements from the image. Printing at original size: 25 mm/s, 10 mm/mV; scaled printing will alter the calibration on paper.

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12-Lead Electrocardiogram Report

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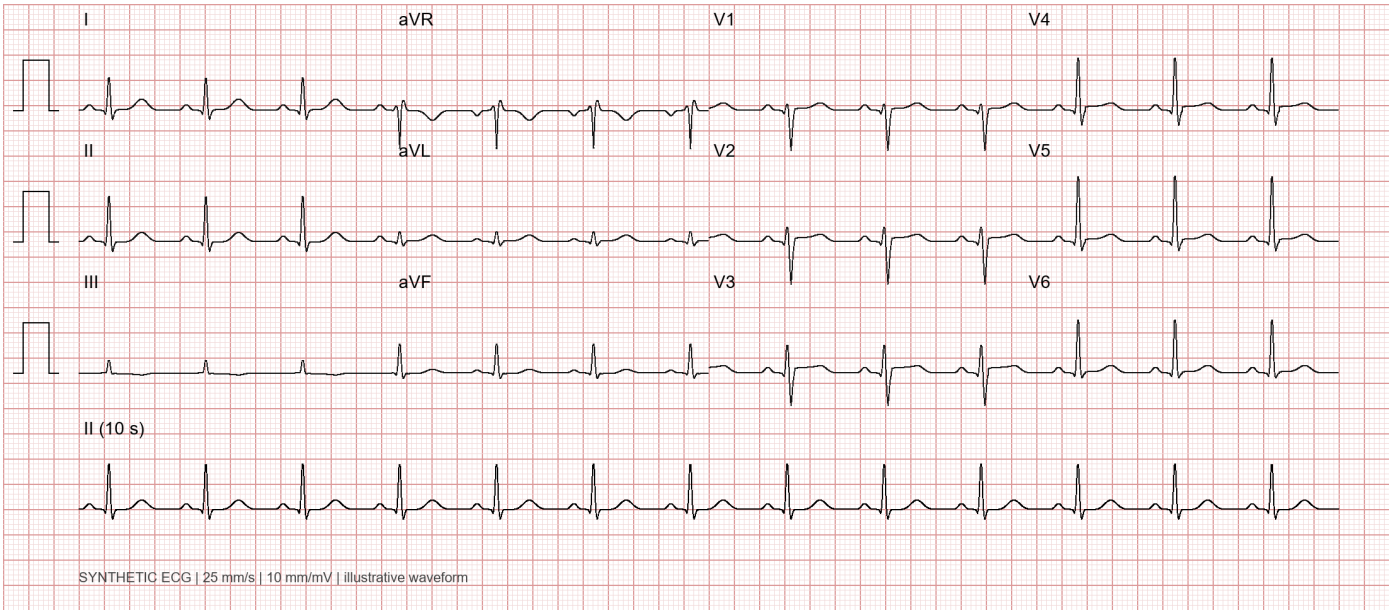
Patient and examination details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex / Age	Male / 58 years
Case ID	SIM-AMI-20260901	Examination date and time	2026-09-01 11:20
Heart rate	78 beats/min	Status	Post-reperfusion

Sinus rhythm. ST elevation in V3 decreased from 4 mm to approximately 1 mm, with marked resolution in the other leads showing elevation. The total amplitude of ST elevation in the precordial leads decreased by >70%. Together with resolution of chest pain and TIMI grade 3 flow, these findings support effective reperfusion.

Synthetic schematic waveforms, not acquired by a device; the quantitative values in this report are specified for the case and must not be replaced by automated image measurements. Print at original size: 25 mm/s, 10 mm/mV; scaled printing will alter the calibration on paper.

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12-Lead Electrocardiogram Report

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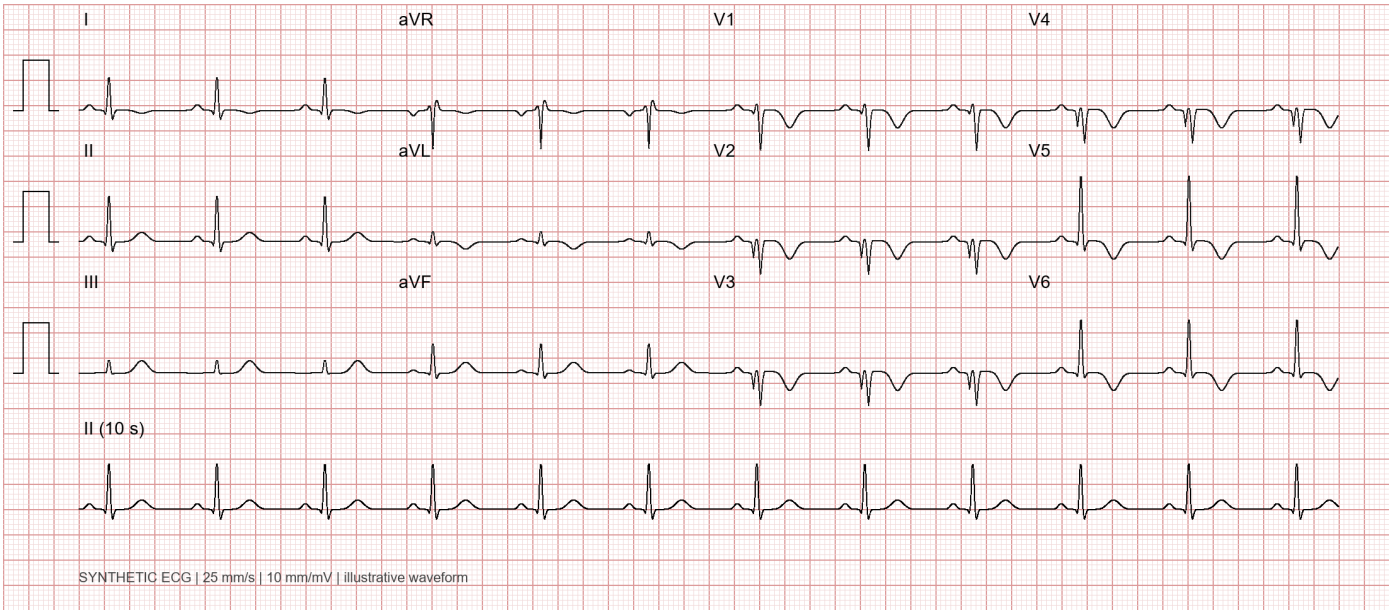
Patient and examination details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex / Age	Male / 58 years
Case ID	SIM-AMI-20260901	Examination time	2026-09-02 07:00
Heart rate	70 beats/min	Phase	Evolving myocardial infarction

Sinus rhythm; pathological Q waves or QS complexes in V2–V4, T-wave inversion in V2–V6, and residual ST elevation ≤0.5 mm. Findings suggest an evolving anterior wall myocardial infarction, with no evidence of new acute ischemia.

Synthetic schematic waveforms, not acquired by a device; quantitative values in this report are specified by the case scenario and must not be replaced by automated measurements from the image. Print at original size: 25 mm/s, 10 mm/mV; scaled printing will alter the calibration on paper.

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12-Lead Electrocardiogram Report

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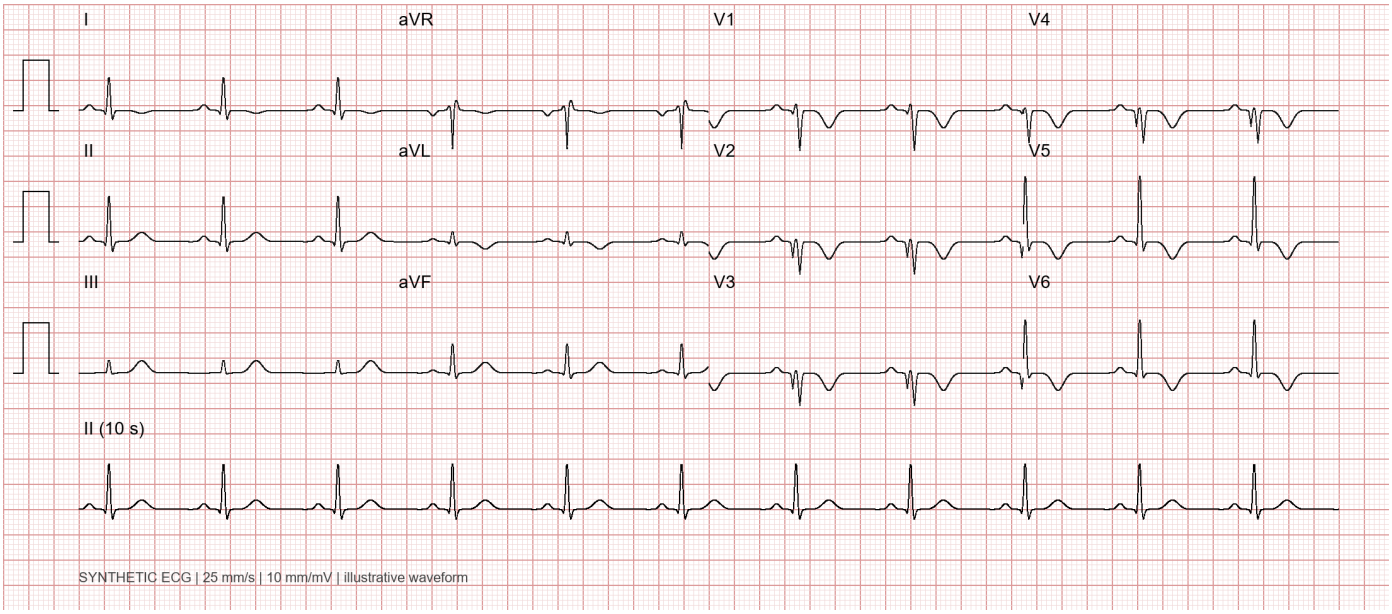
Patient and examination details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex / Age	Male / 58 years
Case ID	SIM-AMI-20260901	Examination date and time	2026-09-06 08:00
Heart rate	66 beats/min	Examination timing	Before discharge

Sinus rhythm; persistent Q waves in V2–V4, T-wave inversion in V2–V6, and no new ST elevation or depression. Findings suggest post-infarction evolution.

Synthetic schematic waveforms, not acquired by a device; quantitative report values are predefined for the case and must not be replaced by automated measurements from the image. Print at original size: 25 mm/s, 10 mm/mV; scaled printing will alter the calibration on paper.

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Echocardiography Report

Medical record photocopy format / Synthetic teaching material

Patient and examination information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Examination time	2026-09-01 12:10
Examination method	Transthoracic echocardiography	Report number	SIM-ECHO-01

Echocardiographic findings

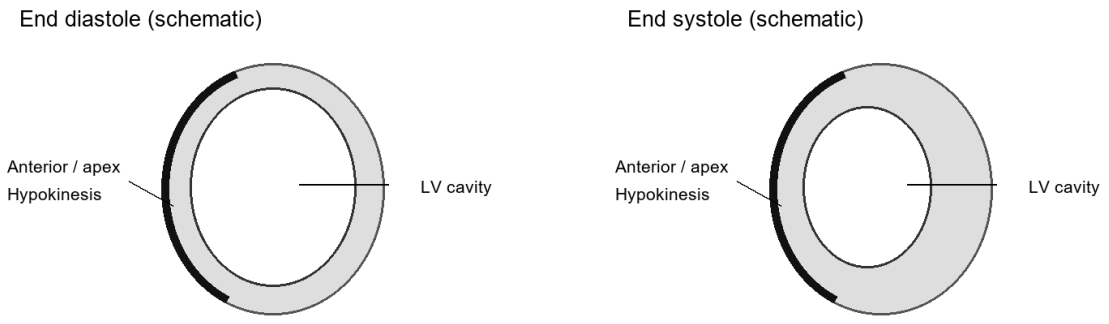
Item	Result	Note / Interpretation
Left ventricular end-diastolic diameter	51 mm	Initial measurement; do not fabricate new values for follow-up examinations
LVEF (biplane Simpson method)	45%	↓ Below this report's lower reference limit for men of 52%
Regional wall motion	Hypokinesis of the mid-to-distal anterior wall and anteroseptum, and the apex	Abnormal; consistent with the left anterior descending artery territory
Right ventricle	Normal size and systolic function on initial examination	Follow-up examination did not suggest any new abnormalities
Valves	Mild mitral regurgitation	Initial finding; follow up in conjunction with the clinical course
Left ventricular thrombus	No definite mural thrombus identified	Routine transthoracic echocardiography has limitations in detection
Other	No ventricular septal rupture, obvious papillary muscle rupture, or pericardial effusion identified	No definite mechanical complications

Echocardiographic impression

Regional wall motion abnormalities in the left anterior descending artery territory of the left ventricle, with mildly reduced global left ventricular systolic function. Correlate with the history of acute anterior wall myocardial infarction.

Examining physician: Simulated Ultrasound Physician A. Reviewing physician: Simulated Ultrasound Physician B. (Both are role identifiers; no actual signatures have been provided.)

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Case LVEF: 45% | Not for volume or EF measurement

Schematic, not acquired ultrasound or DICOM. Illustrates regional wall motion.

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Echocardiography Report

Medical record copy format / Synthetic teaching material

Patient and examination information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Examination time	2026-09-06 08:30
Examination method	Transthoracic echocardiography	Report number	SIM-ECHO-02

Echocardiographic findings

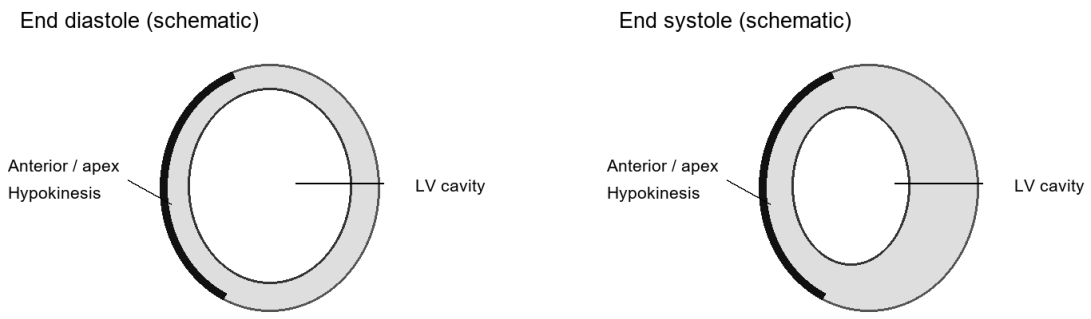
Item	Result	Note / Interpretation
Left ventricular end-diastolic diameter	No new value specified	Measured at the initial examination; no new value fabricated for the follow-up examination
LVEF (biplane Simpson method)	50%	↓ Below this report's male lower reference limit of 52%
Regional wall motion	Hypokinesis of the mid-to-distal anterior wall and anteroseptum, and the apex	Abnormal; improved on follow-up compared with the previous examination
Right ventricle	Normal size and systolic function at the initial examination	No new abnormality indicated on follow-up
Valves	Mild mitral regurgitation	Initial finding; follow-up in the clinical context
Left ventricular thrombus	No definite mural thrombus seen	Routine transthoracic echocardiography has limitations in detection
Other	No ventricular septal perforation, obvious papillary muscle rupture, or pericardial effusion seen	No definite mechanical complications

Echocardiographic impression

Regional wall motion abnormalities in the left anterior descending coronary artery territory, with mildly reduced global left ventricular systolic function. Improved compared with September 1, but the LVEF of 50% remains below this report's male lower reference limit; it cannot be described as completely normal.

Examining physician: Simulated Ultrasound Physician A. Reviewing physician: Simulated Ultrasound Physician B (both are role identifiers; no actual signatures have been provided).

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Case LVEF: 50% | Not for volume or EF measurement

Schematic, not acquired ultrasound or DICOM. Illustrates regional wall motion.

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Chest X-ray Examination Report

Medical record photocopy format / Synthetic teaching material

Patient and examination information

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00	Examination time	2026-09-01 13:00
Report time	13:30	Examination number	SIM-CXR-0901

Examination Technique

Bedside frontal chest radiograph. The examination was performed after PCI and did not delay coronary reperfusion.

Imaging Findings

No obvious consolidation or signs of pulmonary edema in either lung. No obvious blunting of the costophrenic angles or pleural effusion. No obvious enlargement of the cardiac silhouette.

Imaging Impression

No definite acute pulmonary consolidation, pulmonary edema, or pleural effusion. A bedside frontal radiograph has limitations in assessing cardiac size and certain abnormalities; correlate with clinical findings.

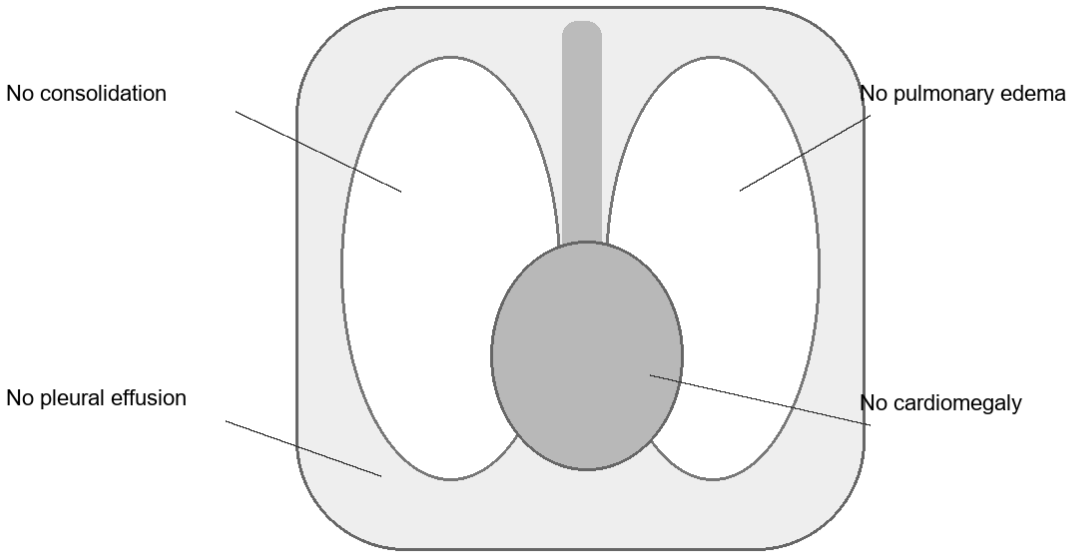
The image is a teaching illustration of the findings described above. A normal chest radiograph cannot be used to exclude acute coronary syndrome, pulmonary embolism, or aortic dissection.

Reporting and review personnel

Role	Designation
Reporting physician	Simulated Radiologist A
Reviewing physician	Simulated Radiologist B

Both are role identifiers; no actual signatures have been provided.

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Report schematic, not a radiograph; not for diagnostic interpretation.

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Coronary Artery Schematic Before and After Intervention

Medical record photocopy format / Synthetic teaching material

Patient and admission details

Field	Value	Field	Value
Name	Zhang Jianguo (fictional)	Sex	Male
Age	58 years	Medical record number	SIM-AMI-20260901
Department	Cardiology / CCU	Admission	2026-09-01 09:20
Discharge	2026-09-07 10:00		

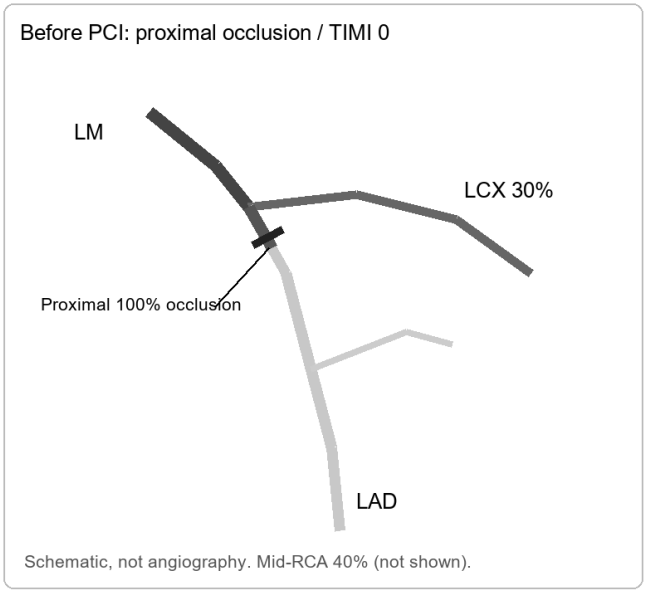
An anatomical schematic corresponding to the operative record; this is not a coronary angiography frame and has no diagnostic imaging validity.

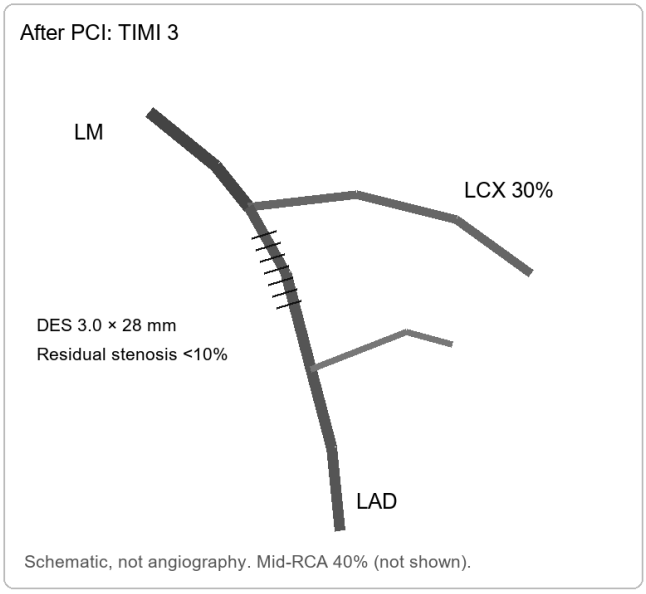
Review points and case specifications

Review point	Case specification
Culprit vessel	Proximal LAD, before the origin of the first diagonal branch
Blood flow changes	TIMI 0 → 2 (after predilation) → 3 (after stenting and optimization)
Stent	3.0×28 mm drug-eluting stent; postdilation of the proximal-to-mid segment with a 3.5 mm noncompliant balloon
Imaging optimization	IVUS minimum stent area approximately 7.1 mm^2, satisfactory apposition, no edge dissection
Nonculprit vessels	30% stenosis in the mid LCX and 40% in the mid RCA; no intervention required during this admission

For a formal assessment of diagnostic imaging interpretation skills, authorized, de-identified real ECG, echocardiography, and coronary angiography data should be added; the synthetic images in this document are suitable for reviewing the clinical logic of the case.

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Basis of Preparation and Data Notes

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Knowledge Base and Guideline Scope

Prepared using the framework in Internal Medicine for the symptoms, electrocardiographic evolution, cardiac biomarkers, and complications of acute myocardial infarction. No textbook edition is specified, and no page numbers or direct quotations have been fabricated. All case values were manually assigned and are not taken from real patients.

1. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for Acute Coronary Syndromes

Used for emergency reperfusion, interventional, and antithrombotic strategies.

2. AHA: Key Points of the 2025 ACS Guideline

For ACS without high bleeding risk, the default duration of dual antiplatelet therapy is at least 12 months, adjusted according to risk.

3. 2023 ESC Guidelines for Acute Coronary Syndromes

Used for the framework for emergency reperfusion in STEMI, inpatient care, and long-term management.

4. ESC: Long-Term Clinical Management After ACS

LDL-C <1.4 mmol/L with a reduction of ≥50%; repeat lipid testing in 4–6 weeks.

5. JACC: 2026 Lipid Management Guideline Portal

Incorporates the secondary prevention framework from the 2026 lipid management update.

Standardization in This Edition

Clarifies that the initial statin dose is not repeated on the same day; lists the maintenance duration of dual antiplatelet therapy separately. Adds synthetic metadata, including laboratory specimens and report times. Discharge LVEF of 50% remains marked ↓. Alert cutoffs for random blood glucose and NT-proBNP are distinguished from diagnostic thresholds. All newly added times are solely for simulated page layout and are not claimed to originate from equipment or a hospital information system.

Limitations of Images and Reference Intervals

The four 12-lead electrocardiograms are programmatically generated schematic signals that conform to paper-speed and amplitude calibration and illustrate ST/T/Q evolution, but are not equivalent to complete physiological electrical signals. The ultrasound, chest radiograph, and coronary artery illustrations use explicitly labeled structural/motion schematics rather than original medical images; they are not suitable for assessing interpretation skills for actual medical images. Laboratory reference intervals and sex-specific thresholds are defined by this simulated laboratory; actual use must follow the applicable equipment, reagents, and institution.

Search/preparation date: 2026-09-10. Citation links are clickable. Guidelines are used to explain the principles of preparation; synthetic patient values must not be treated as empirical evidence from guidelines.

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Clinical Expert Review Form

Medical Record Copy Format / Synthetic Teaching Material

Case ID: SIM-AMI-20260901 Version: v1.0 Reviewing expert: _____ Date: _____

Review Dimensions and Suggested Focus Areas

Review dimension	Suggested focus areas	Comments
History and diagnosis	Chest pain characteristics, differential diagnosis, localization, and evidence for type 1 myocardial infarction	
Timeline	ECG, medication administration, guidewire, balloon, stent, and postprocedural report	
Laboratory consistency	Reference intervals, ↑↓, units, trends over time, and sampling status	
ECG	Whether ST/T/Q evolution is consistent with LAD occlusion and reperfusion	
Interventional treatment	Culprit vessel, stent dimensions, IVUS, anticoagulation, and procedural rationale	
Inpatient management	Cardiac function assessment, screening for complications, and medication selection and doses	
Discharge plan	DAPT, lipid lowering, rehabilitation, follow-up testing, and emergency response education	
Suitability for teaching	Limitations of synthetic images, case difficulty, and decision points suitable for training	

Overall assessment: □ Suitable for teaching □ Use after revision □ Case requires reconstruction (please check manually)

Page Numbers Requiring Revision, Issues, and Recommendations

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